Finite Volume method

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FDM to FVM

Suppose you have a physical domain , the first approach to FDM is to divide the domain into a computational mesh and define a series of points. The function is evaluated at the node(grid) points. NO concern regarding the variation of the function between these grid points.This makes sense because the derivates evaluated using finite differences , they use the value at the grid points only .

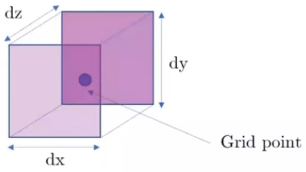
Building block of FVM

For FVM we consider the same grid points , so let this be an arbitrary grid point somewhere in the domain



In the finite volume method we consider a volume about the grid point and the grid point is the centroid of that volume.

Finite volume



IN 3D the dimensions of that finite volume could be Δx,Δy,Δz

A common technique for FVM is to take the computationl mesh from FDM and then substitute a finite volume around everypoint



So we will have 4 finite volume around 4 interior grid points here .



The boundary points will also have a finite volume but the volume itself will be halfed because it wouldnt fit the entire domain. This formulation of FVM is useful when you have boundary condition such that boundary values are defined.



So the interior points will have a full finite volume but for the boundary points they may have to do some special treatments.

Another way to formulate the finite volume methode is without mesh grid points at boundaries. This formulation is useful when we have a boundary FLUX rather than boundary values.



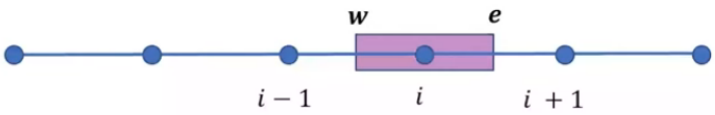
Methodology of FVM
Consider Pure diffusion equation :

$$\frac{d^2T}{dx^2} = 0 \tag{1}$$

For FDM we approach the problem by taylor series and we consider a general grid point indexed by i and from index we mean the count of that point.



After applying Finite central differencing we obtained that the temperature at i is just the average of its neighbours $T_i = \frac{1}{2}(T_{i+1} + T_{i-1})$ - this was the methodology of FDM where in order to arrive at an iterative formula we use taylor series expansion.
However in the case of finite volume , the idea is to integrate the governing equation over the finite volume.



Here my grid point of interest is i and it has two neighbours $i+1$ and $i-1$.The finite volume is centered around grid point i,it will lie between $i+1$ and $i-1$.The boundaries of the control volume are positioned midway between adjacent nodes.Where the boundaries cross the grid ,we call thoses faces w-**west face** and e-**east face**.
Then we integrate over the finite volume :

$$\int \frac{d^2T}{dx^2}\, dV = 0 \tag{2}$$

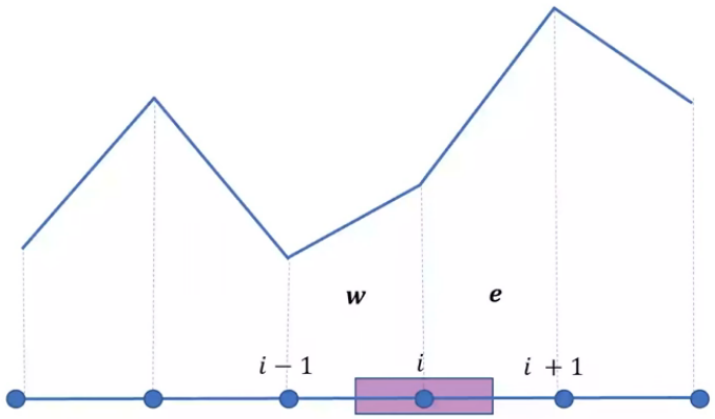
Since we are in one dimensional system we can conviently replace dV by dx

$$\int_w^e \frac{d^2T}{dx^2}\, dx = 0 \tag{3}$$

$$\left.\frac{dT}{dx}\right|_e - \left.\frac{dT}{dx}\right|_w = 0 \tag{4}$$

DEPENDING on how we assume the variation of function between two grid points,we can numerically approximate the derivative
Between i and $i+1$ we can have any type of variation for the function,if the function is linear , we will one kind of derivative and if the function is a different polynomial, it could be another type of derivative.So depending on how the function might vary between two grid points these derivatives might change.

A good starting point is to assume **piecewise linear variation** between two grid points.

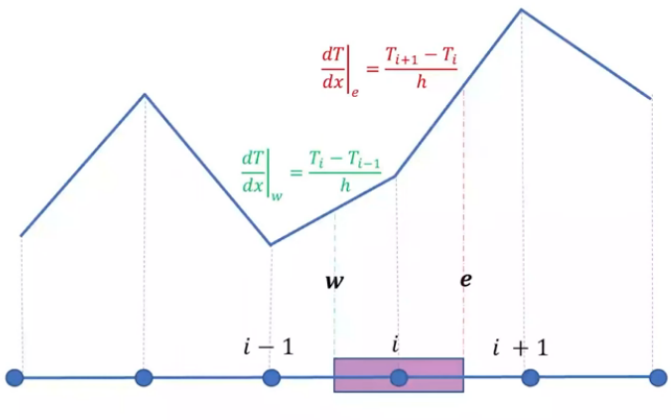


Here we can safely say that

$$\left.\frac{dT}{dx}\right|_e = \frac{T_{i+1} - T_i}{\delta x} \tag{5}$$

, where δx is the width of the finite volume.This formula is nothing but the central differencing from taylor series expansion.Similarly we can write the derivative of the west face

$$\left.\frac{dT}{dx}\right|_w = \frac{T_i - T_{i-1}}{\delta x} \tag{6}$$


$\left.\frac{dT}{dx}\right|_e = \frac{T_{i+1} - T_i}{h}$
$\left.\frac{dT}{dx}\right|_w = \frac{T_i - T_{i-1}}{h}$
w
e
$i-1$
i
$i+1$

Once we substitute back into EQN(4) we can easily see that

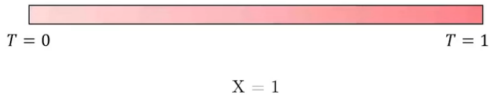
$$\frac{T_{i+1} - 2T_i + Ti - 1}{\delta x} \tag{7}$$

Rearranging we get

$$T_i = \frac{1}{2}(T_{i+1} + T_{i-1}) \tag{8}$$

Piecewise linear scheme will result in a similar result as that of central differences.

Numerical example



X = 1

Given: The physics of the situation is governed by the differential equation of pure diffusion:

$$\frac{d^2T}{dx^2} = 0$$

Initially: $T(t = 0, x < 1) = 0$ and $T(t = 0, x = 1) = 1$ (Initial conditions)
At any other instant: $T(t = t, x = 0) = 0$ and $T(t = t, x = 1) = 1$ (Boundary conditions)

Objective: To calculate the temperature distribution, $T(x)$, at equilibrium.

Framework

Primary variables we will need for the code :

1. Position array x
2. 2 Temperature arrays (one old and 1 new)
3. Scalers ; N= Number of grid points , h = grid spacing ,ϵ is the error threshold

Algorithm

Variable definition and initialization

while the error is larger than the required error
    for all the grid points
        perform the iterations to calculate new values of T
        calculate the revised error
    end
end

Basically there is another way to solve the equation , which is the MATRIX method rather than iterative method

1. First the discretised eqn of the form of EQN(8) must be set up at each of the nodal points in order to solve the problem , the goal is to obtain a system of linear algebraic equations.

2. Chapter 7 from CFD VERSTEEG book offers the solution of equations in details. but just note that their is a DIRECT way (matrix) and IN-Direct way (iterative) to solve the problem.

Bibliography

[1] H. K. Versteeg and W. Malalasekera, *An Introduction to Computational Fluid Dynamics: The Finite Volume Method*, 2nd ed., Pearson, 2007.

[2] F. M. White, *Fluid Mechanics*, 8th ed., McGraw-Hill, 2016.